Ali Ayati

+1-979-344-9829 | ali.a@tamu.edu | https://ali-ayati.com

aliayati | cpt9m0 | SA Ayati

College Station, TX

Objective

PhD student in Computer Engineering specializing in Systems Security and AI-Driven Analysis. Seeking a Software Engineering or Site Reliability Engineering internship to apply research in kernel-level instrumentation, distributed systems, and scalable ML pipelines to real-world infrastructure challenges.

Education

- **Texas A&M University** — *January 2023 – Expected Dec 2027*
  PhD in Computer Engineering (3.60/4.00): Thesis Advisor Prof. Dr. Marcus Botacin — College Station, TX
  Thesis: Access Control Is All You Need
- **Iran University of Science and Technology** — *September 2017 - February 2022*
  B.S. in Computer Engineering (3.50/4.00): Thesis Advisor Dr. Saeed Parsa — Tehran, IR
  Thesis: CodART: An Automated Refactoring System (https://github.com/m-zakeri/CodART)

Skills

Programming Languages	Python, C, C++, Java
Databases	SQL, MongoDB, Redis, Amazon RDS
Systems & Security	Windows Kernel-Mode Driver Framework, PE Files, Antlr, Ghidra, Yara
Backend & Web	Django, HTML, CSS, FastAPI, Flask
Machine Learning	Graph Analysis, Random Forest, PyTorch, PEFT (LoRA/QLoRA)
DevOps & Tooling	Git, CI/CD, Docker, Linux

Publications

PR=Peer-Reviewed,

* Major Contribution

P.R.[1] ***Seyyed Ali Ayati**, Jin Hyun Park, Yichen Cai, Marcus Botacin (2025). **Making Acoustic Side-Channel Attacks on Noisy Keyboards Viable with LLM-Assisted Spectrograms' "Typo" Correction** In *The 19th USENIX WOOT Conference on Offensive Technologies (WOOT '25)* – https://www.usenix.org/conference/woot25/presentation/ayati

Research Experience

Research Assistant — *January 2024 – Present*
Botacin's Lab, Texas A&M University (College Station, TX)

- **Systems Architecture & Kernel Development:** Architected a real-time Endpoint Detection and Response (EDR) system for Windows, implementing C/C++ kernel-mode drivers for low-latency system call interception and high-performance threat monitoring.
- **Algorithm Optimization:** Designed and benchmarked 7+ prevention algorithms, utilizing data mining techniques to reduce False Positive Rates (FPR) to less than 1.0% while maintaining robust prevention capabilities.
- **Scalable Data Analysis:** Engineered a parallelized analysis pipeline to process 60M+ kernel trace events, utilizing hierarchical tree structures and anomaly detection algorithms to identify discriminative access patterns at scale.
- **Automation Infrastructure:** Developed an automated evaluation framework to validate prevention metrics (TPR, FPR, Precision) across 6,000+ samples, enabling continuous testing of algorithm configurations against diverse malware datasets.
- **Policy Enforcement:** Built a proactive access control generation system that prevents malware execution (rather than just detection) by enforcing blocking rules on malicious patterns before runtime execution.

Teaching Assistant Experience

Texas A&M University

- Software Security (CSCE 413) — Spring 2025, Spring 2026
- Data Analytics Cybersecurity (CSCE 704) — Fall 2025
- Operating Systems (CSCE 410/611) & Design and Analysis of Algorithms (CSCE 411) — Summer 2024, 2025
- Senior Capstone Design (CSCE 482/483) — Spring 2024, Fall 2024

Iran University of Science and Technology

- Artificial Intelligence (CSCE 420) — Spring 2022
- Digital Systems Design (ECEN 248) — Fall 2021
- Compiler Design (CSCE 434) — Fall 2020, Spring 2021

Recent Projects

EchoCrypt: High-Performance Side-Channel Analysis Framework *Aug 2024 – Mar 2025*
https://github.com/Botacin-s-Lab/EchoCrypt

- **ML Systems Engineering:** Developed a modular acoustic side-channel attack framework integrating Vision Transformers (ViT, Swin) and LLMs, optimizing inference pipelines to achieve 90%+ recovery accuracy in noisy environments.
- **Model Optimization (LoRA):** Implemented Low-Rank Adaptation (LoRA) fine-tuning for LLaMA models, reducing computational memory requirements by approx. 90% while retaining near-GPT-4 contextual correction capabilities.
- **Data Pipeline Automation:** Built end-to-end data augmentation and evaluation pipelines using PyTorch and CUDA, enabling reproducible, large-scale benchmarks across diverse audio datasets (Zoom, Phone).

Interactive Malware Feature Engineering Lab (Full-Stack & DevOps) *Oct 2024 – Nov 2025*
lab.ali-ayati.com

- **Production Deployment:** Architected and deployed a containerized web application (Docker, Nginx, Gunicorn), configuring reverse proxies to handle concurrent user sessions for real-time model training.
- **Backend Scalability:** Engineered a Django-based backend to manage budget-constrained Random Forest training jobs, processing 33+ static PE features with optimized scikit-learn pipelines.
- **Interactive Systems:** Designed a gamified interface that provides real-time feedback on feature importance, bridging complex malware analysis data with user-friendly visualization.

Certifications

- **GitHub Foundations –** Issued by GitHub (Credentials: dfa3484e-ab86-4bda-84da-173c6b37d452)
- **Data Parallelism: How to Train Deep Learning Models on Multiple GPUs –** Issued by NVIDIA (Credentials: 438cdf74cd0c427cbccbbec604822469)
- **Fundamentals of Accelerated Data Science –** Issued by NVIDIA (Credentials: pVK2Ck4OQEiVGgbVnyZvAg)
- **GRAD Aggies Professional Development Certificate –** Issued by Graduate and Professional School at Texas A&M University (Credentials: 8SUspTJzQGiCs1PJ4tIaOQ)